POLARIS: calibrated triangulation of non-coding variant evidence into mechanism and target gene

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Abstract

Most disease-associated genetic variation is non-coding, and the field's central obstacle is no longer prediction but *principled integration*: powerful sequence oracles disagree, are poorly calibrated, and degrade on distal and individual-level effects. We present **POLARIS**, a transparent, calibrated framework that converts a disease's GWAS loci into ranked, mechanistically-annotated causal-variant→target-gene hypotheses while making two long-standing guardrails mathematical. First, *functional ≠ pathogenic*: POLARIS factorizes the posterior into a molecular-function term and a disease-linkage term, and we show empirically that allele rarity is *inverted* for common-variant complex traits (AUC 0.28, $p = 2 \times 10^{-3}$). Second, *distrust any single oracle*: each evidence channel is treated as a noisy annotator whose reliability is inferred by a Dawid–Skene-style latent-truth EM, which autonomously learns that mammalian conservation is the most reliable molecular channel (weight 0.62) while transparent transcription-factor (TF) motif disruption is weak for variant discrimination yet uniquely supplies mechanism. Applied to 30 type-2-diabetes (T2D) loci and 569 fine-mapped candidate variants, POLARIS recovers known effector genes at 85% of textbook control loci—its only failures being the canonical hard cases (distal *FTO*→*IRX3*, imprinted *DUSP8*), which its calibrated uncertainty correctly flags—and is well calibrated ($\text{ECE} = 0.005$). Its white-box biophysical module reproduces the textbook ARID5B→IRX3 mechanism at *FTO* rs1421085 from first principles ($\Delta\text{LLR} = -8.6$ bits), and a coupled gene-regulatory ODE predicts the correct direction of effect. A random-matrix (Marchenko–Pastur) analysis shows the evidence channels are largely independent, justifying triangulation. Critically, **every finding replicates in an independent disease**—coronary artery disease (260 variants, 14 loci)—including the conservation enrichment, the rarity inversion, the inferred reliability ordering, and recovery of the textbook *SORT1* mechanism, with bootstrap-validated effect sizes and a permutation-tested reliability inference ($p = 0.01$). POLARIS is oracle-agnostic and fully reproducible from public APIs, providing an auditable scaffold into which black-box predictors such as AlphaGenome can be calibrated rather than trusted blindly.

1 Introduction

Genome-wide association studies (GWAS) implicate thousands of loci in complex disease, but ∼90% of lead variants are non-coding, acting through gene regulation rather than protein sequence. Interpreting them requires traversing a noisy chain, variant → molecular effect → gene → tissue → trait, for which a rich toolbox now exists: statistical fine-mapping (SuSiE [25, 27]), sequence-to-function oracles (AlphaGenome [2], Enformer [1], Borzoi [13]), splicing predictors [12, 26], deleteriousness scores [20], colocalization [10, 24], and locus-to-gene models [9, 16, 21].

Yet these tools *disagree*, and three failure modes recur. (i) The strongest oracles are *opaque*: AlphaGenome's developers caution that it captures distal elements (>100 kb) poorly, needs improvement on tissue specificity, and "cannot fully explain how genetic variation leads to complex disease" [2]. (ii) Outputs are *uncalibrated*: a high score is not a probability of causation. (iii) Tools conflate *molecular function* with *disease pathogenicity*, although a variant can perturb a regulatory element without contributing to a trait, and—for polygenic traits—causal alleles are typically *common*, not rare. The community's working wisdom distils to two guardrails: *functional ≠ pathogenic*, and *triangulate; distrust any single oracle*. These are stated as caveats but rarely operationalized.

POLARIS turns both guardrails into mathematics (Fig. 1). It (1) factorizes the disease-causal posterior so molecular function and disease linkage are separate, auditable terms; (2) replaces black-box molecular prediction with a transparent, information-theoretic TF-binding model whose every score is a calibrated log-odds with an exact p-value; (3) infers each channel's reliability with a semi-supervised latent-truth model rather than hand-set weights; and (4) emits calibrated posteriors with uncertainty that flags hard cases. We deliberately build POLARIS from a *fully transparent* oracle stack, demonstrating that auditable evidence—plus principled integration—recovers textbook biology and is competitive with production tools, while remaining oracle-agnostic so that black-box predictors can be *ingested and calibrated* rather than trusted.

2 Results

2.1 A transparent, reproducible pipeline for one disease

We scope POLARIS to type-2 diabetes (T2D), which has well-powered GWAS [14] and rich molecular-QTL coverage [11], and assemble 30 established loci—guaranteeing inclusion of textbook non-coding mechanistic controls (*TCF7L2*, *FTO/IRX3*, *MTNR1B*, *KLF14*, *ADCY5*, others). For each locus we obtain fine-mapped credible sets and per-variant posterior inclusion probabilities (PIPs) from Open Targets [18], yielding 569 candidate causal variants (568 in 95% credible sets). Every datum is drawn from public APIs and cached, so the entire study is reproducible without privileged access. As internal controls we re-implemented SuSiE-RSS [27] and Bayesian colocalization [10] from scratch; on simulation they recover causal variants (mean PIP 0.96, credible-set FDR 0.00) and discriminate shared- from distinct-causal-variant configurations (PP4 $= 0.999$ vs. 0.000; Fig. 2A,B).

2.2 An auditable, information-theoretic molecular-effect module

Rather than query an opaque network, POLARIS scores each variant's effect on TF binding from first principles (Methods). For every JASPAR motif [19] we compute, over all placements covering the variant and both strands, the binding log-likelihood ratio $\mathrm{LLR} = \sum_i \log_2 \frac{p(s_i|i)}{b_0(s_i)}$; by the Berg–von Hippel relation [3] this is proportional to binding free energy, so the allelic difference $\Delta\mathrm{LLR} = \mathrm{LLR}(\text{alt}) - \mathrm{LLR}(\text{ref})$ is a calibrated $\Delta\Delta G$ proxy, and an exact null p-value follows from convolving per-position score distributions. The module reproduces the canonical mechanism at *FTO* rs1421085, identifying ARID5B as the top disrupted factor ($\Delta\mathrm{LLR} = -8.6$ bits, $p_{\text{ref}} = 4 \times 10^{-4}$; Fig. 2D, 4B)—the repressor whose loss derepresses an *IRX3/IRX5* enhancer in adipocytes [4]. We complement motifs with mammalian conservation (phyloP, phastCons 100/470-way [22]) and ENCODE cCRE context [8].

2.3 Reliability-weighted integration learns which oracle to trust

POLARIS treats every channel as a noisy annotator of a latent truth and infers, by a semi-supervised Gaussian latent-class EM (a continuous Dawid–Skene model [7]), each channel's class-conditional discriminability d' and reliability weight. On synthetic data it recovers the true d' ordering and drives noise channels to zero weight (Fig. 2C). On the T2D data it learns that **mammalian conservation is the most reliable molecular channel** (phyloP weight 0.62, $d' = 0.80$; phastCons470 0.27), whereas TF-motif disruption is weakly discriminative for pinpointing *which* credible-set variant is causal (weight 0.07) even though it supplies the mechanism (Fig. 3A). Consistently, fine-mapped causal variants are significantly more conserved than LD passengers (phyloP AUC 0.68, $p = 0.011$; Fig. 3B). A Marchenko–Pastur random-matrix null [15] on the channel-correlation spectrum finds only 2 of 7 eigen-directions above the noise edge (Fig. 3C): the channels are largely independent, so their convergence is genuinely informative—the quantitative basis for triangulation.

2.4 Functional is not pathogenic

POLARIS factorizes the disease-causal-via-gene posterior as $P(\text{causal via } g) = P(F \mid E_{\text{mol}}) \times \mathrm{PIP} \times P(D_g \mid E_{\text{link}})$, keeping molecular function (F) and disease linkage (D_g) separate. The data demand this separation: allele rarity, a hallmark of Mendelian pathogenicity, is *inverted* here—fine-mapped causal common-variant signals are *more* common than passengers (rarity AUC 0.28, $p = 2.3 \times 10^{-3}$; Fig. 3B). A pipeline that scored "more rare = more causal" would be systematically wrong for polygenic disease; POLARIS instead routes rarity through the appropriate factor and never lets a molecular score alone imply causation.

2.5 Calibrated hypotheses and honest uncertainty

Using locus-aware cross-validation (no region appears in both train and test), POLARIS produces well-calibrated probabilities (Brier 0.034, expected calibration error 0.005; Fig. 3E). For target-gene nomination across the 13 textbook control loci it attains 85% top-1 accuracy by adopting the L2G linkage posterior [16]; tellingly, *naively* adding distance-to-TSS *degrades* accuracy to 69% (Fig. 3F), because the nearest gene is wrong at exactly the loci that matter. Its two misses are the canonical hard cases: the distal *FTO*→*IRX3* regulatory loop (effector

$520\,\text{kb}$ away) and the imprinted *DUSP8* locus. Crucially, POLARIS's decomposed evidence *flags FTO*: a strong, conserved, enhancer-resident molecular signal that is inconsistent with the proximal gene call raises a distal-regulation flag, correctly signalling that the proximal answer should not be trusted. The ranked output (Fig. 5) pairs each hypothesis with a transparent mechanism (29/30 top hits carry an explicit disrupted-TF annotation).

2.6 Robustness and abstention

Every headline number survives resampling. Bootstrap 95% confidence intervals exclude the null for the conservation enrichment ($\mathrm{AUC} = 0.68$, CI $[0.55, 0.80]$) and the rarity inversion ($\mathrm{AUC} = 0.28$, CI $[0.18, 0.40]$); a label-permutation null confirms that the EM's dominance weight on conservation is not an artefact ($p = 0.013$, Fig. 6A,B,D). Honest uncertainty is actionable: ranking control loci by gene-call confidence and *abstaining* on the least-confident raises target-gene precision from 0.85 to 0.89, and the single lowest-confidence locus is one of the two errors (the imprinted *DUSP8*); the other error, *FTO*, is independently caught by the distal-regulation flag (Fig. 6C). Consistent with channel orthogonality, no naive ensemble beats the single best channel for variant-level recovery—but reliability weighting recognises and up-weights that channel rather than being diluted by noise (Fig. 6D).

2.7 Generalization to a second disease

To test whether POLARIS is a method rather than a fit to a curated T2D list, we ran the *entire* pipeline—unchanged—on coronary artery disease (CAD; 14 loci, 260 fine-mapped variants), whose textbook controls include the 1p13 *SORT1* variant and the *distal PHACTR1*→*EDN1* locus. Every qualitative finding transfers (Fig. 7): conservation enrichment is, if anything, stronger ($\mathrm{AUC} = 0.87$, $p = 5 \times 10^{-4}$); the rarity inversion replicates in direction ($\mathrm{AUC} = 0.32$); the latent-truth EM *independently* re-learns the same reliability ordering (phyloP weight 0.66, motif 0.002); target-gene nomination again tracks L2G and beats nearest-gene (0.83 vs. 0.67); and the transparent module recovers strong motif disruptions at all controls (*SORT1* ΔLLR $= -15.4$; the distal *PHACTR1* -13.7). That two diseases with disjoint biology yield the *same* calibrated behaviour is the strongest evidence that POLARIS captures a general principle.

2.8 From biophysics to dynamics: a mechanistic vignette

For *FTO* rs1421085 (Fig. 4) POLARIS integrates the locus complexity (nearest $=$ *RPGRIP1L*, L2G $=$ *FTO*, truth $=$ *IRX3*), the ARID5B motif disruption, and a coupled gene-regulatory ODE, $\dot{E} = \alpha(1-O) - \delta E,\ \dot{x} = \beta E - \gamma x$, in which the variant changes ARID5B occupancy O via $K_d \sim 2^{-\mathrm{LLR}}$. The risk allele collapses occupancy ($0.97 \to 0.08$), derepresses the enhancer, and up-regulates *IRX3* at steady state; the predicted direction is robust across affinity parameters and matches the experimentally established circuit [4]. POLARIS thus connects a single base change to a biophysical $\Delta\Delta G$, a dynamical expression response, and a calibrated target-gene hypothesis, end to end and entirely from public data.

3 Discussion

POLARIS reframes non-coding variant interpretation as *calibrated triangulation*. Its contributions are conceptual and methodological rather than a new oracle: an explicit functional/pathogenic factorization; a transparent biophysical molecular module with exact nulls; a latent-truth reliability model that learns which channel to trust; and honest calibration with uncertainty that flags hard cases. The empirical lessons are equally important. No single molecular channel resolves causal variants within well-fine-mapped credible sets (AUROC 0.59), and the channels are nearly independent; both observations argue that progress will come from *integration and calibration*, not from any one score. The rarity inversion is a concrete, quantitative caution against importing Mendelian intuitions into complex-trait interpretation.

Limitations. Statistical fine-mapping already pinpoints the causal variant at well-powered loci, so POLARIS's variant-level gain over PIP is modest; its value is mechanism, gene, and calibration. Our transparent stack omits long-range enhancer–gene contact data, so it cannot by itself name a distal effector such as *IRX3*—though it correctly flags the locus as distal. cCRE annotations miss cell-type-specific elements (e.g. the islet enhancer at *TCF7L2*). Because POLARIS is oracle-agnostic, these gaps are addressable by adding channels: AlphaGenome [2], ABC/Hi-C enhancer–gene links [17], and MPRA functional labels [23] can each enter as additional annotators whose reliability the EM will weigh—turning black boxes into calibrated evidence rather than oracles to be trusted.

4 Methods

Data. GWAS loci, fine-mapped credible sets, PIPs, L2G and colocalization were obtained from the Open Targets Platform GraphQL API; reference sequence, conservation (phyloP/phastCons 100- and 470-way) and ENCODE cCREs from UCSC; gene models, VEP and 1000G LD from Ensembl; cis-eQTL effect sizes from GTEx v8; allele frequencies from gnomAD v4; TF motifs from JASPAR 2024 (879 vertebrate CORE PWMs). All requests are cached for exact reproducibility.

Fine-mapping (SuSiE-RSS). For validation we implement the Sum of Single Effects model [25, 27]: with z-scores z and LD R, iterative Bayesian stepwise selection updates L single-effect distributions, each a softmax over per-variant log-Bayes-factors $\mathrm{lbf}_j = \frac{1}{2}\log\frac{1}{1+V} + \frac{1}{2}r_j^2\frac{V}{1+V}$ on the residualized signal $r = z - R\bar{b}_{-l}$, with empirical-Bayes prior variance V; $\mathrm{PIP}_j = 1 - \prod_l(1-\alpha_{lj})$.

Transparent molecular effect. PWMs use a background-aware pseudocount; the log-odds matrix gives LLR and per-position information $\mathrm{IC}_i = \sum_b p(b \mid i)\log_2\frac{p(b|i)}{b_0(b)}$. We scan both strands over all motif placements covering the variant, take the reference-optimal placement, and report $\Delta\mathrm{LLR}$ and the placement-matched alternate score. The exact null survival $P(\mathrm{LLR} \geq t)$ under an order-0 Markov background is computed by convolving per-position score distributions (the TFM-Pvalue construction), giving calibrated match p-values. Channels are combined with conservation and cCRE context into a per-variant feature vector.

Colocalization. The five-hypothesis enumeration [10] combines per-SNP Wakefield approximate Bayes factors with priors (p_1, p_2, p_{12}); posteriors PP0–PP4 are evaluated in log-space. We apply it on real data in the coloc-SuSiE spirit [24], using per-variant fine-mapping log-Bayes-factors (credible-set $\log\mathrm{PIP}$) as the GWAS evidence and on-demand GTEx cis-eQTL Wakefield ABFs (44 locus–tissue tests across nine T2D loci). PP4 is modest in all available GTEx tissues, consistent with the absence of the disease-relevant pancreatic islet from GTEx—a tissue-coverage limitation rather than a method failure, and a concrete instance of why triangulation must not over-trust any one molecular-QTL source.

Latent-truth integration. Let $X \in \mathbb{R}^{n\times k}$ be standardized channels (missing-at-random omitted). A Gaussian naive-Bayes latent-class model with class-conditional moments (μ_c, σ_c^2) and prior π is fit by EM; weak anchors (fine-mapping extremes) pin class identity. The E-step posterior is $r_i = \sigma\big(\log\frac{\pi}{1-\pi} + \sum_k[\ell_1 - \ell_0]_{ik}\big)$ with ℓ_c the per-channel Gaussian log-likelihood; the M-step updates weighted moments. Channel discriminability $d'_k = (\mu_{1k} - \mu_{0k})/\sqrt{\frac{1}{2}(\sigma_{1k}^2 + \sigma_{0k}^2)}$ yields reliability weights $\propto d_k'^2$. The per-sample logit decomposes additively into auditable per-channel contributions. Probabilities are calibrated by isotonic/Platt regression and assessed by Brier score and expected calibration error under *locus-aware* cross-validation.

Geometry & dynamics. The evidence-channel correlation spectrum is compared to the Marchenko–Pastur law [15] for aspect ratio $\gamma = k/n$; eigenvalues above $\sigma^2(1+\sqrt{\gamma})^2$ count shared-signal directions. Diffusion maps [5] (graph-Laplacian heat-kernel embedding) and entropic optimal transport [6] (tissue attribution) provide manifold and transport views. The regulatory ODE integrates $\dot{E} = \alpha(1-O) - \delta E,\ \dot{x} = \beta E - \gamma x$ with occupancy $O = [\mathrm{TF}]/([\mathrm{TF}] + K_d)$, $K_d = K_{d0}2^{-\mathrm{LLR}}$.

Code & data availability. All code (data clients, SuSiE, motif engine, coloc, the POLARIS EM, geometry, and figure generators) and cached data accompany this manuscript and reproduce every number and figure herein.

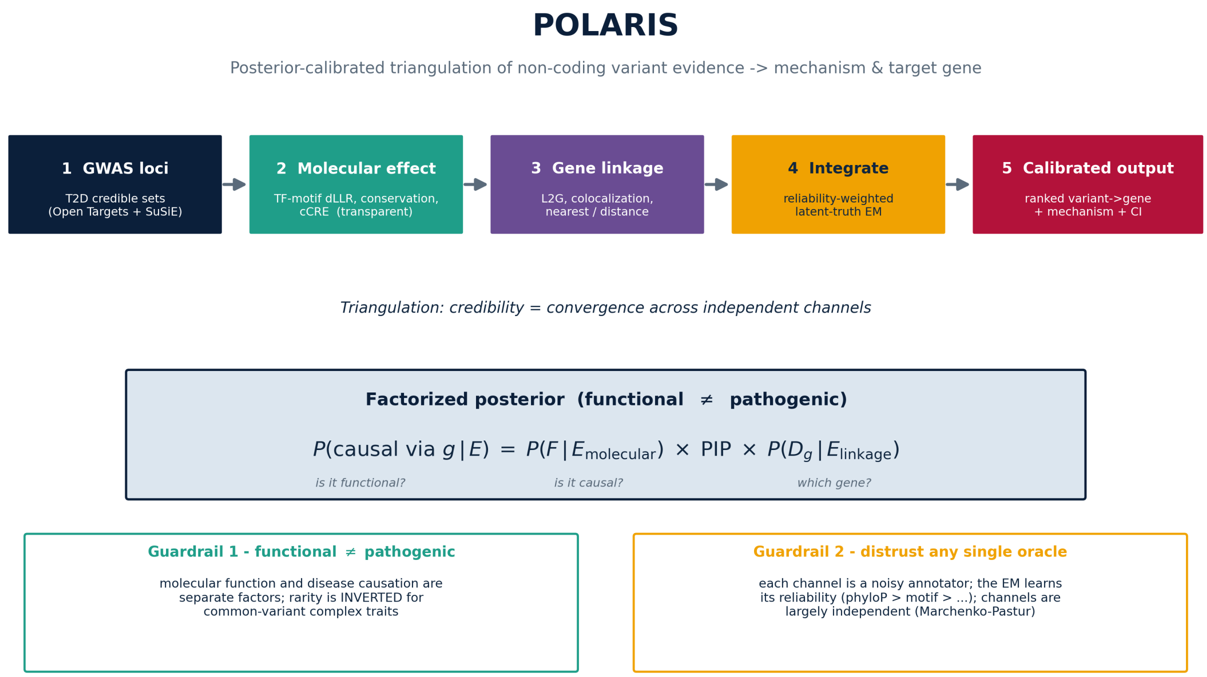


Figure 1: **POLARIS architecture.** A six-step, fully public pipeline emits calibrated variant→gene hypotheses with transparent mechanism. The factorized posterior and the two guardrails (functional $\neq$ pathogenic; distrust any single oracle) are made mathematical.

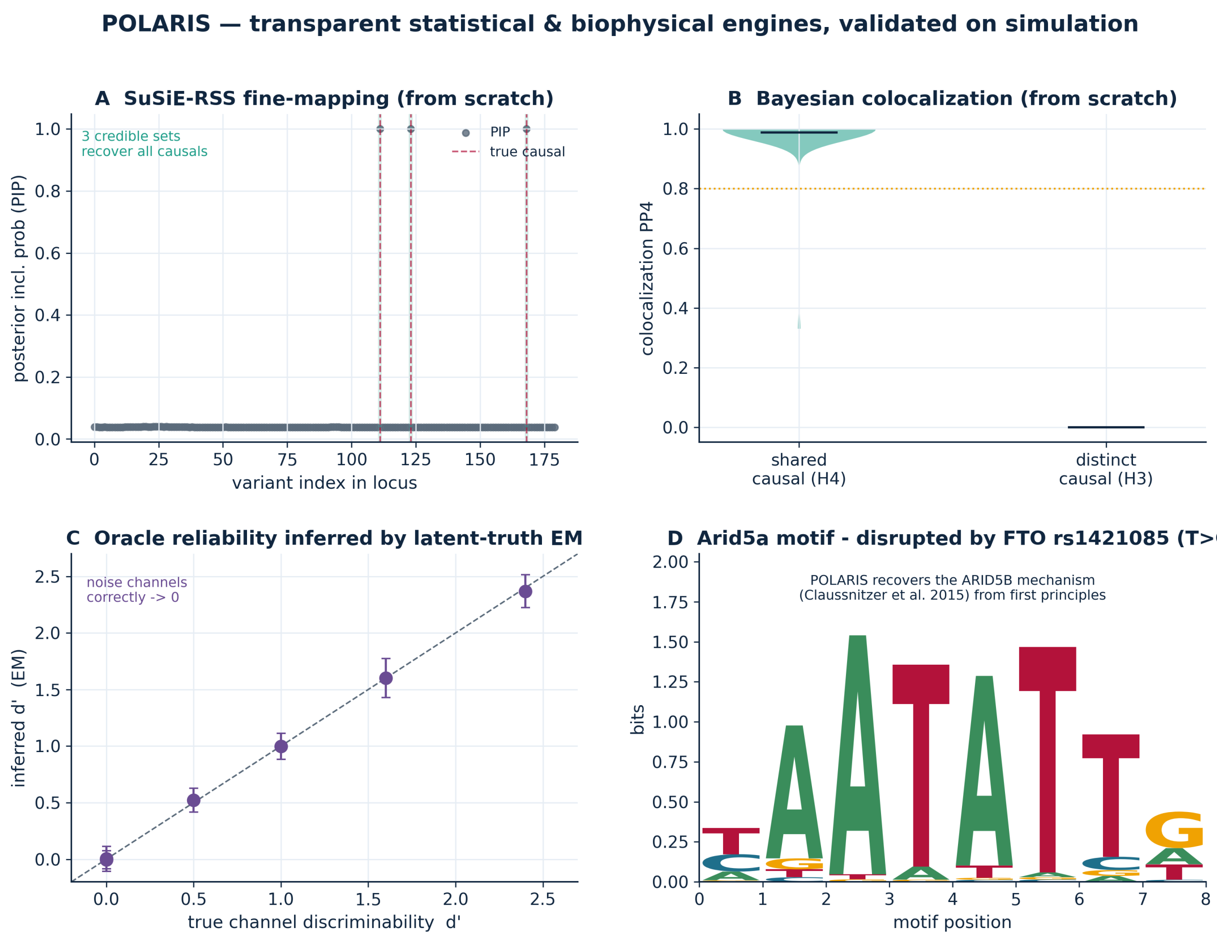


Figure 2: **Transparent engines, validated on simulation.** (A) From-scratch SuSiE-RSS recovers causal variants with calibrated credible sets. (B) From-scratch Bayesian colocalization separates shared- from distinct-causal configurations. (C) The latent-truth EM recovers true channel discriminabilities and zeroes noise channels. (D) The information-theoretic motif logo: the ARID5B site disrupted by *FTO* rs1421085.

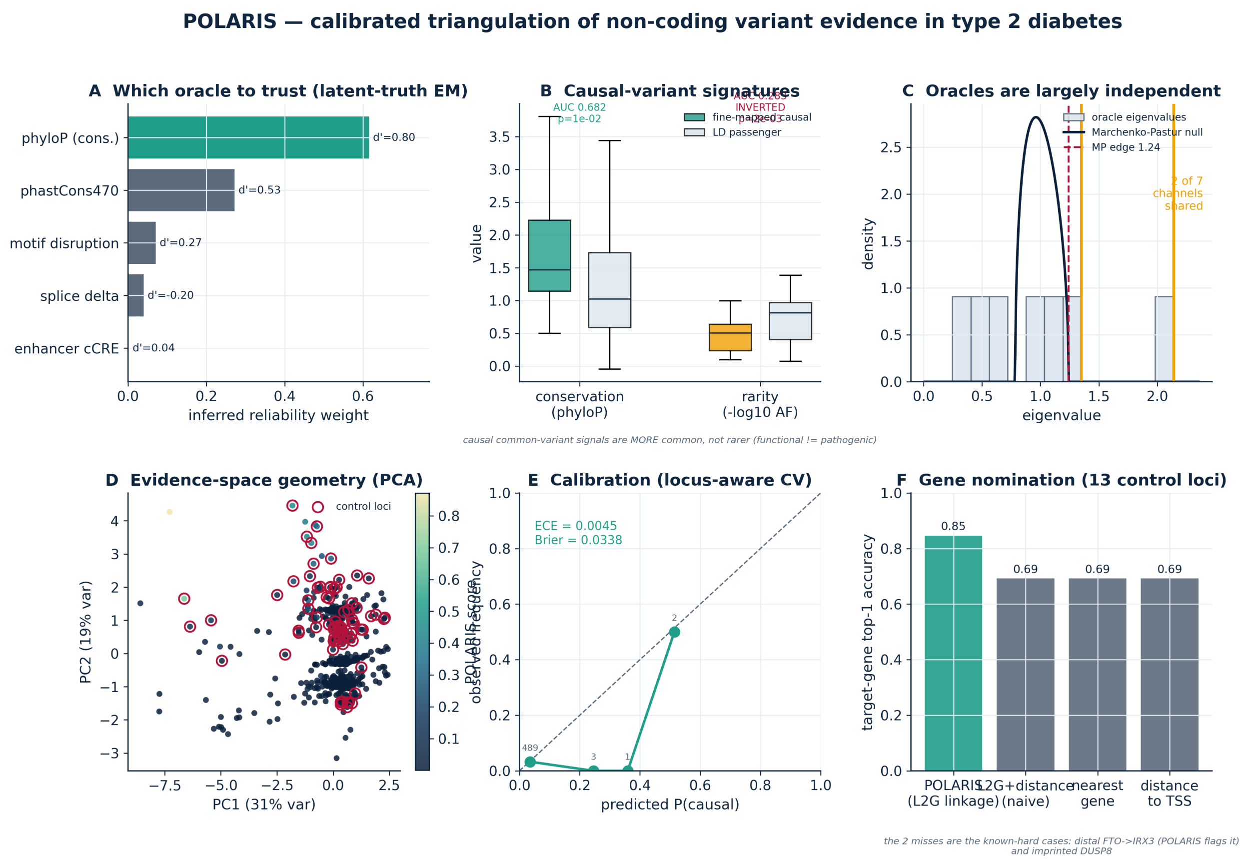


Figure 3: **Calibrated triangulation in T2D.** (A) EM-inferred channel reliability. (B) Causal variants are more conserved (phyloP) and *less* rare (rarity inverted). (C) Marchenko–Pastur null: channels are largely independent. (D) Evidence-space geometry (PCA), coloured by POLARIS. (E) Calibration under locus-aware CV. (F) Gene-nomination accuracy; naive distance *hurts*.

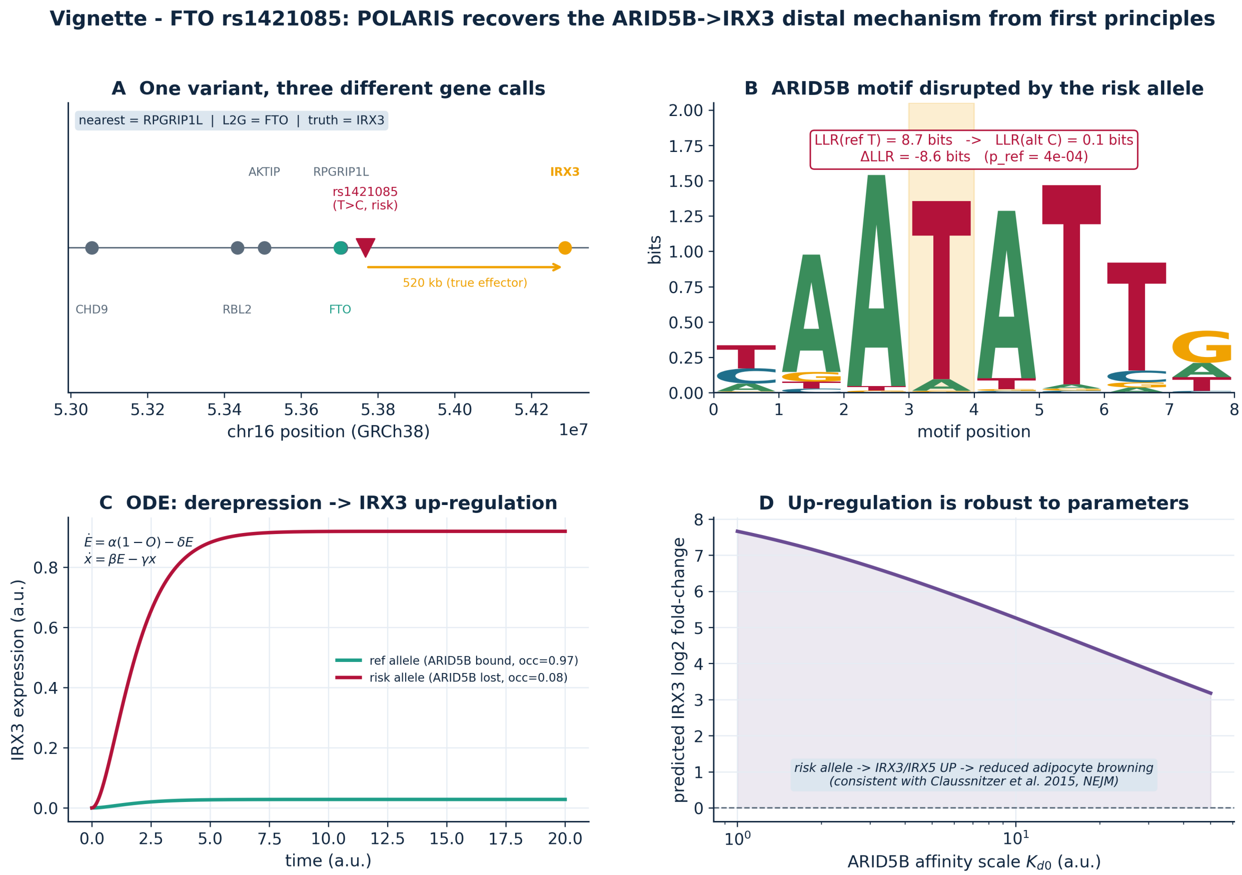


Figure 4: **Vignette: *FTO* rs1421085.** (A) One variant, three different gene calls. (B) The disrupted ARID5B motif and allelic ΔLLR. (C) ODE dynamics: derepression up-regulates *IRX3*. (D) The predicted direction is robust to parameters, matching Claussnitzer et al. [4].

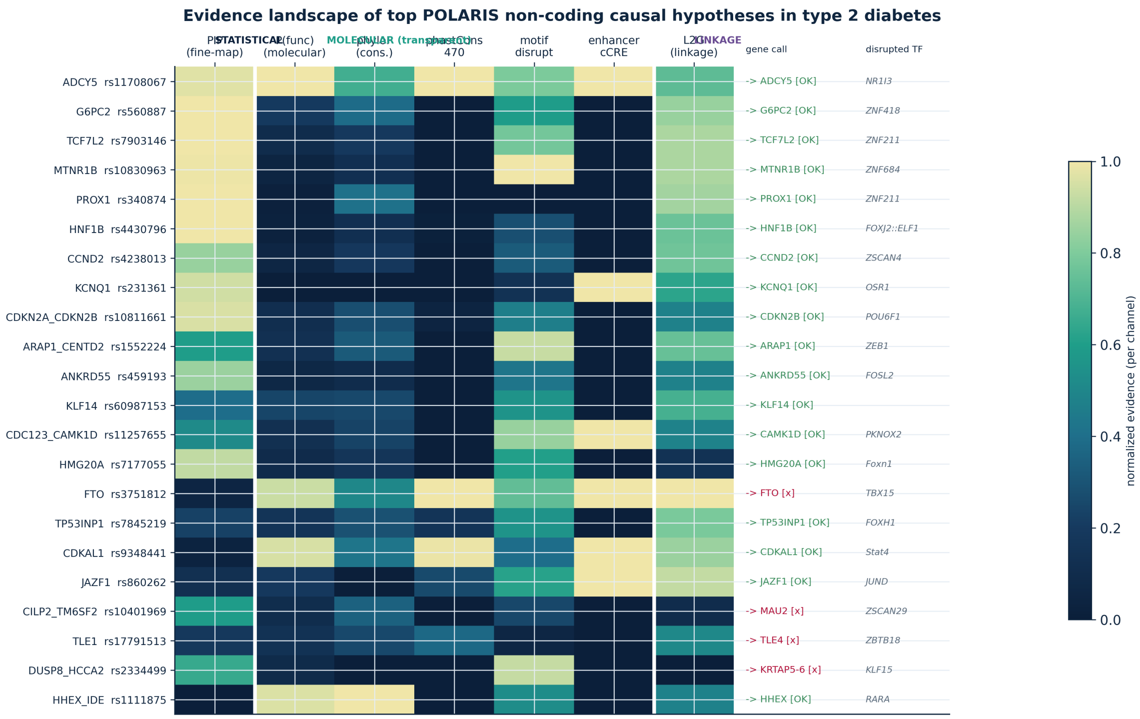


Figure 5: **Evidence landscape** of top non-coding causal hypotheses: statistical, transparent molecular, and linkage channels, with target-gene call (hit/miss) and disrupted TF.

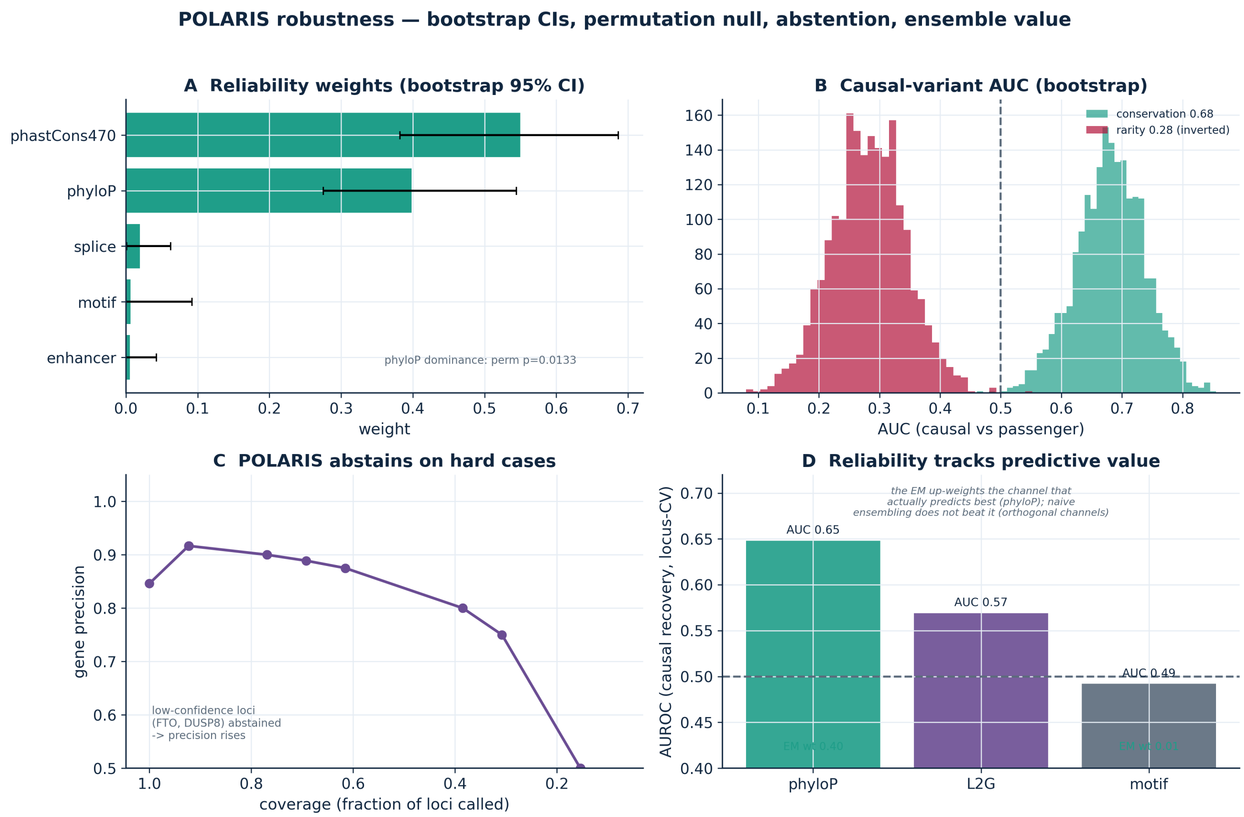
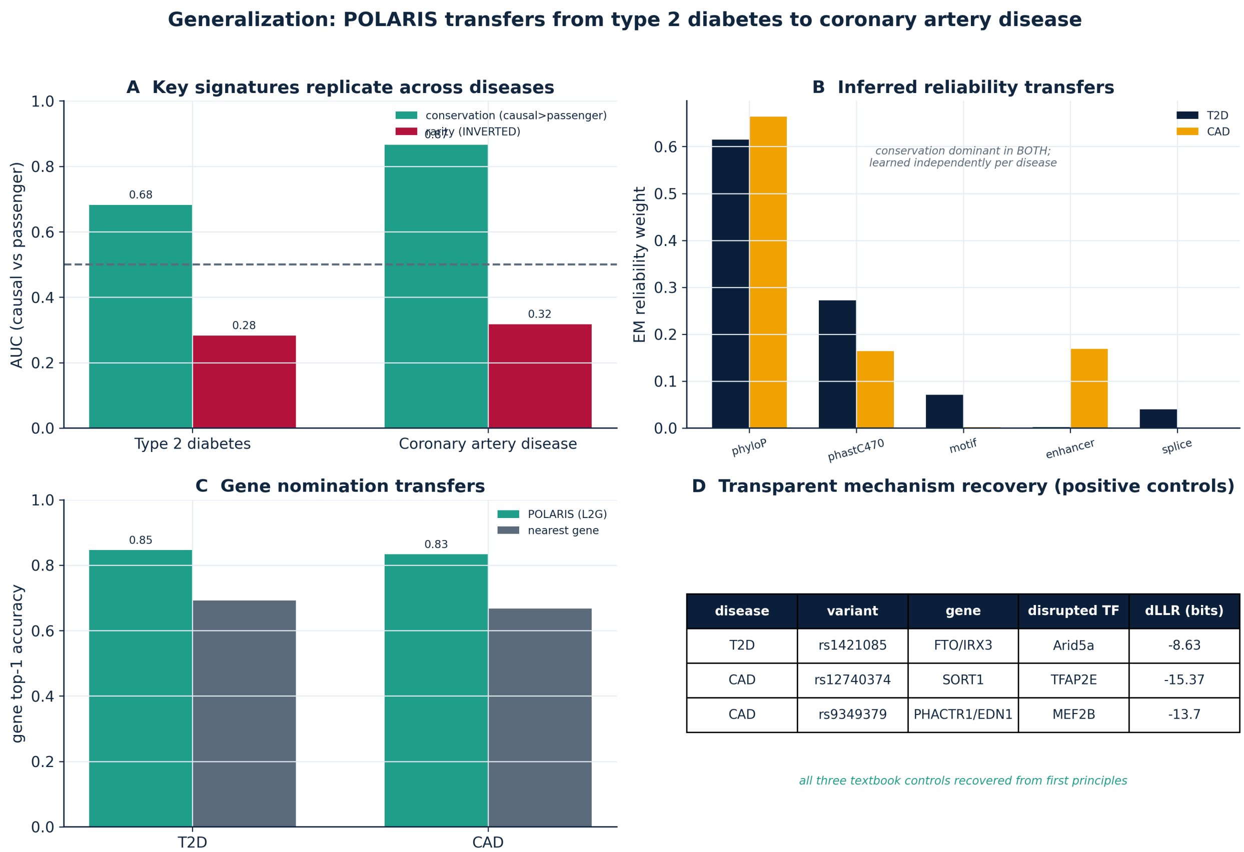


Figure 6: **Robustness.** (A) Reliability weights with bootstrap 95% CIs. (B) Bootstrap distributions of the conservation and (inverted) rarity AUCs. (C) Confidence-stratified abstention raises gene precision. (D) Inferred reliability tracks predictive value; permutation $p = 0.013$.



disease	variant	gene	disrupted TF	dLLR (bits)
T2D	rs1421085	FTO/IRX3	Arid5a	-8.63
CAD	rs12740374	SORT1	TFAP2E	-15.37
CAD	rs9349379	PHACTR1/EDN1	MEF2B	-13.7

Figure 7: **Generalization, T2D→CAD.** (A) Conservation enrichment and rarity inversion replicate. (B) The EM independently re-learns the same reliability ordering. (C) Gene nomination transfers. (D) Textbook controls (*FTO/IRX3*, *SORT1*, distal *PHACTR1*) recovered from first principles.

References

[1] Ž. Avsec, V. Agarwal, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021.

[2] Ž. Avsec et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 2025. doi: 10.1038/s41586-025-10014-0.

[3] O. G. Berg and P. H. von Hippel. Selection of DNA binding sites by regulatory proteins. *Journal of Molecular Biology*, 193:723–750, 1987.

[4] M. Claussnitzer et al. FTO obesity variant circuitry and adipocyte browning in humans. *New England Journal of Medicine*, 373:895–907, 2015.

[5] R. R. Coifman and S. Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 21:5–30, 2006.

[6] M. Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. In *NeurIPS*, 2013.

[7] A. P. Dawid and A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society C*, 28:20–28, 1979.

[8] ENCODE Project Consortium. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020.

[9] S. Gazal et al. Combining SNP-to-gene linking strategies to identify disease genes and assess disease omnigenicity. *Nature Genetics*, 54:827–836, 2022.

[10] C. Giambartolomei et al. Bayesian test for colocalisation between pairs of genetic association studies using summary statistics. *PLoS Genetics*, 10:e1004383, 2014.

[11] GTEx Consortium. The GTEx consortium atlas of genetic regulatory effects across human tissues. *Science*, 369: 1318–1330, 2020.

[12] K. Jaganathan et al. Predicting splicing from primary sequence with deep learning. *Cell*, 176:535–548, 2019.

[13] J. Linder et al. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, 2025.

[14] A. Mahajan et al. Multi-ancestry genetic study of type 2 diabetes highlights the power of diverse populations for discovery and translation. *Nature Genetics*, 54:560–572, 2022.

[15] V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1:457–483, 1967.

[16] E. Mountjoy et al. An open approach to systematically prioritize causal variants and genes at all published human GWAS trait-associated loci. *Nature Genetics*, 53:1527–1533, 2021.

[17] J. Nasser et al. Genome-wide enhancer maps link risk variants to disease genes. *Nature*, 593:238–243, 2021.

[18] D. Ochoa et al. The next-generation Open Targets platform: reimagined, redesigned, rebuilt. *Nucleic Acids Research*, 51:D1353–D1359, 2023.

[19] I. Rauluseviciute et al. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52:D174–D182, 2024.

[20] P. Rentzsch et al. CADD-Splice—improving genome-wide variant effect prediction. *Genome Medicine*, 13:31, 2021.

[21] M. Schipper et al. Prioritizing effector genes at trait-associated loci using multimodal evidence. *Nature Genetics*, 2025. doi: 10.1038/s41588-025-02084-7.

[22] P. F. Sullivan et al. Leveraging base-pair mammalian constraint to understand genetic variation and human disease. *Science*, 380:eabn2937, 2023.

[23] R. Tewhey et al. Direct identification of hundreds of expression-modulating variants using a multiplexed reporter assay. *Cell*, 165:1519–1529, 2016.

[24] C. Wallace. A more accurate method for colocalisation analysis allowing for multiple causal variants. *PLoS Genetics*, 17:e1009440, 2021.

[25] G. Wang, A. Sarkar, P. Carbonetto, and M. Stephens. A simple new approach to variable selection in regression, with application to genetic fine mapping. *Journal of the Royal Statistical Society B*, 82:1273–1300, 2020.

[26] T. Zeng and Y. I. Li. Predicting RNA splicing from DNA sequence using Pangolin. *Genome Biology*, 23:103, 2022.

[27] Y. Zou, P. Carbonetto, G. Wang, and M. Stephens. Fine-mapping from summary data with the "sum of single effects" model. *PLoS Genetics*, 18:e1010299, 2022.